

Inventory performance: \$ value, turns and days of supply

Part 5 · Evaluating performance — solution report with data, method, results and charts

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1. Inventory performance: \$ value, turns and days of supply

Lecture worked examples

Turns = $1,000,000 / 500,000 = 2$ Days of supply = $6,000 / 200 = 30$ days
 $\rightarrow 360/30 = 12$ turns

2

Turns (COGS example)

30 days

Days of supply (unit example)

12 per year

Equivalent turns

Division benchmark

Average inventory is the mean of the opening and four quarter-end balances (a better estimate than a single year-end snapshot, which can be window-dressed).

6.05

Company-wide turns

60 days

Company-wide days of supply

Automotive Components (36.0 turns)

Best division

Automotive Components (-4.0)

Worst vs benchmark

\$42.4 M

Total \$ inventory (Q4)

\$0.93 M/yr

Carrying-cost gap vs benchmarks

\$ value of inventory held (year-end snapshot)

Balance-sheet inventory = RM + WIP + FG

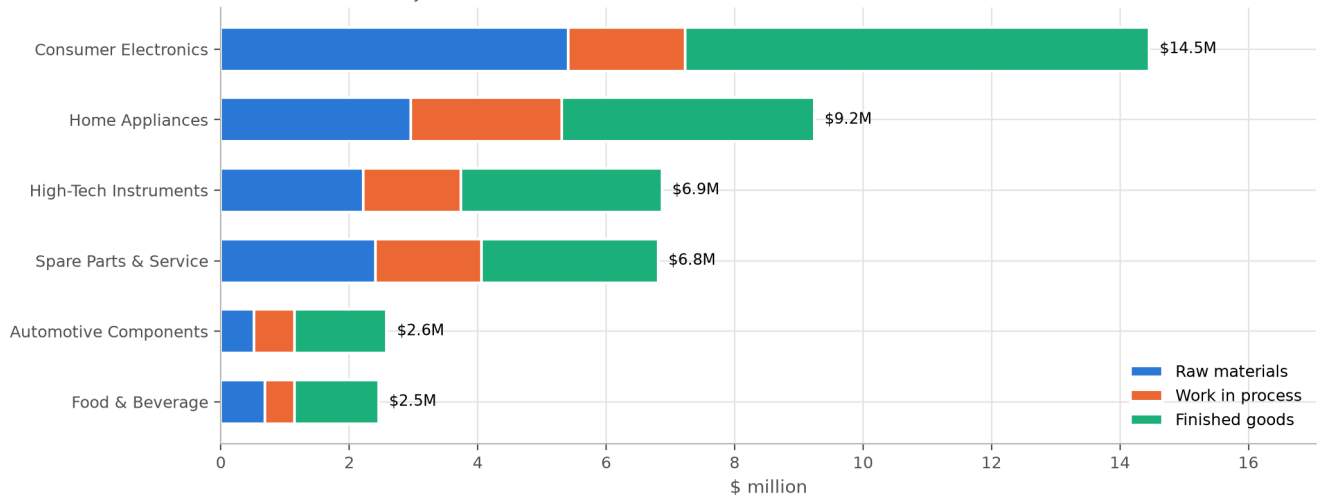


Figure 1.1 — Year-end inventory by division and stage.

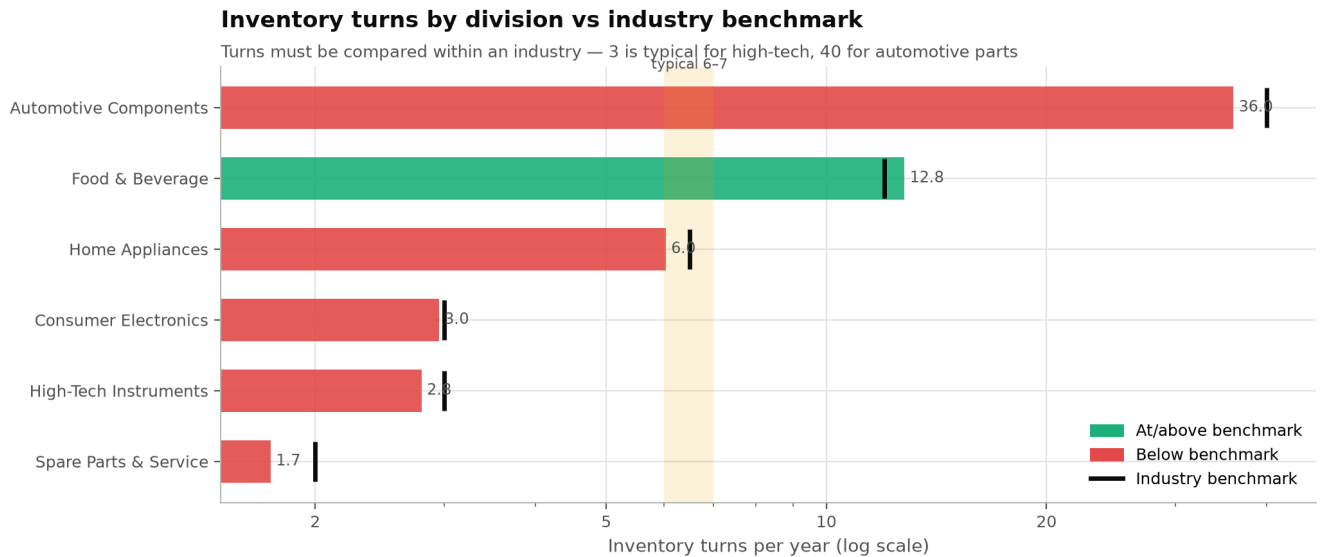


Figure 1.2 — Turns = annual COGS / average inventory.

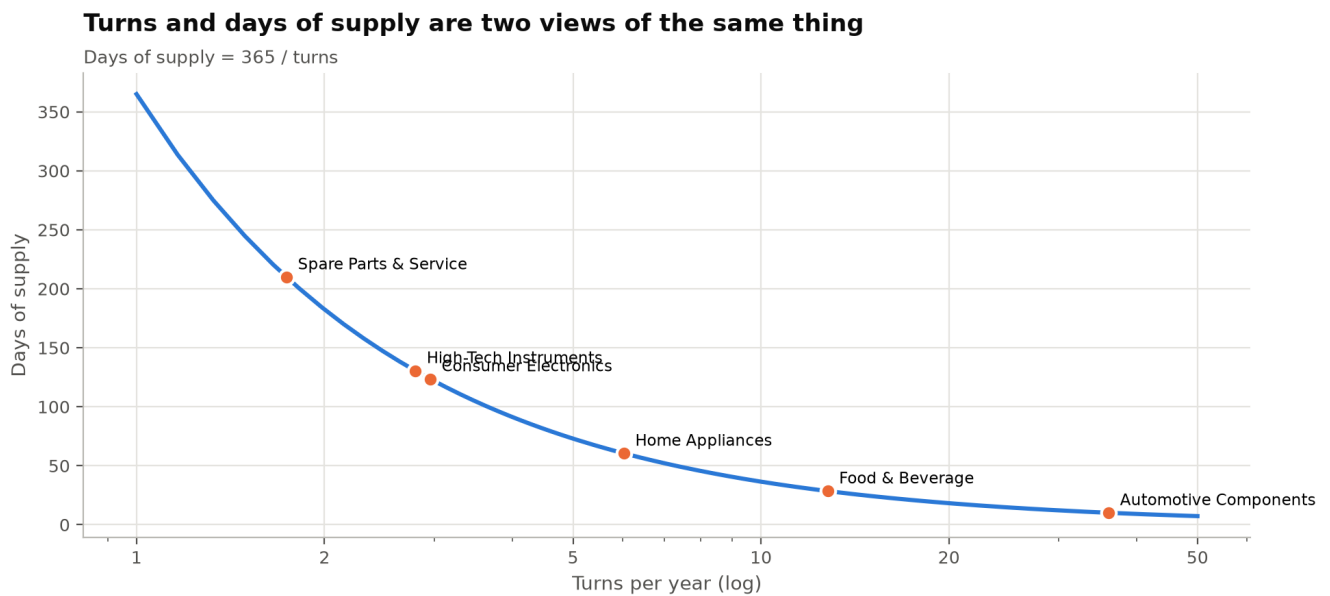


Figure 1.3 — The hyperbola links turns and days of supply.

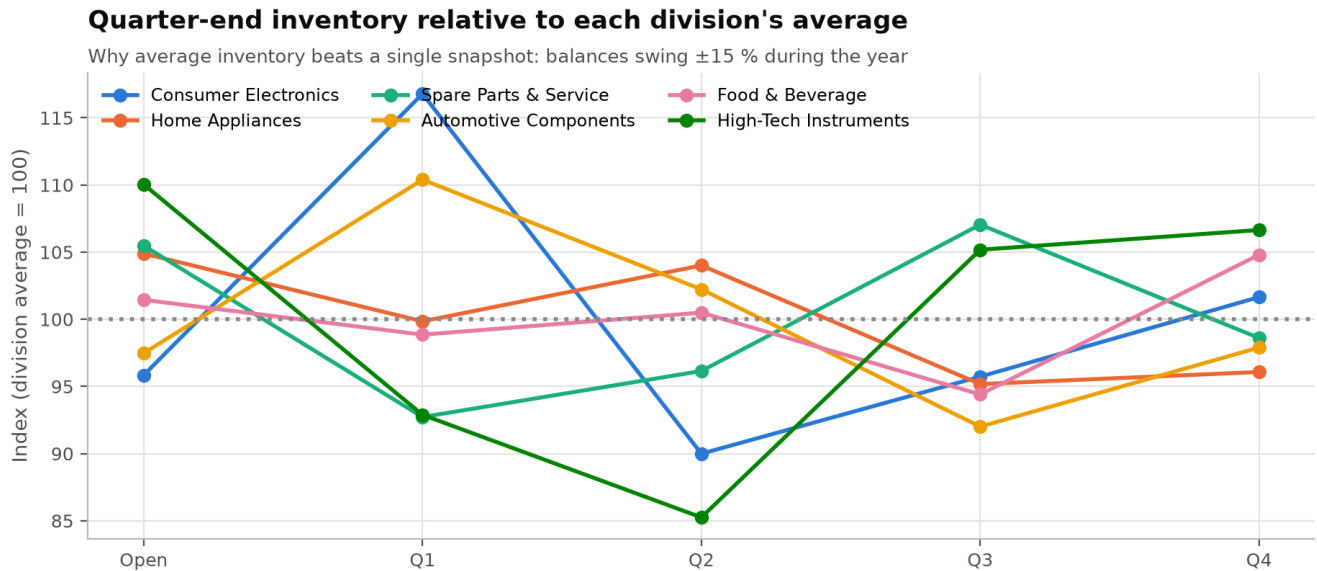


Figure 1.4 — Quarter-end balances indexed to the division's average.

Division turnover analysis.

division	annual cogs	avg inventory	year end inventory	turns	turns on year end	days of supply	industry benchmark turns	turns gap vs benchmark	excess carrying cost
Consumer Electronics	\$42,000,000	\$14,222,563	\$14,457,599	2.95	2.91	124	3.0	-0.05	\$82,348
Home Appliances	\$58,000,000	\$9,617,652	\$9,240,231	6.03	6.28	61	6.5	-0.47	\$256,993
Spare Parts & Service	\$12,000,000	\$6,904,519	\$6,808,137	1.74	1.76	210	2.0	-0.26	\$334,672
Automotive Components	\$95,000,000	\$2,636,740	\$2,581,215	36.03	36.80	10	40.0	-3.97	\$96,844
Food & Beverage	\$30,000,000	\$2,346,218	\$2,458,859	12.79	12.20	29	12.0	+0.79	-\$56,899
High-Tech Instruments	\$18,000,000	\$6,439,082	\$6,867,108	2.80	2.62	131	3.0	-0.20	\$162,460

Item-level days of supply

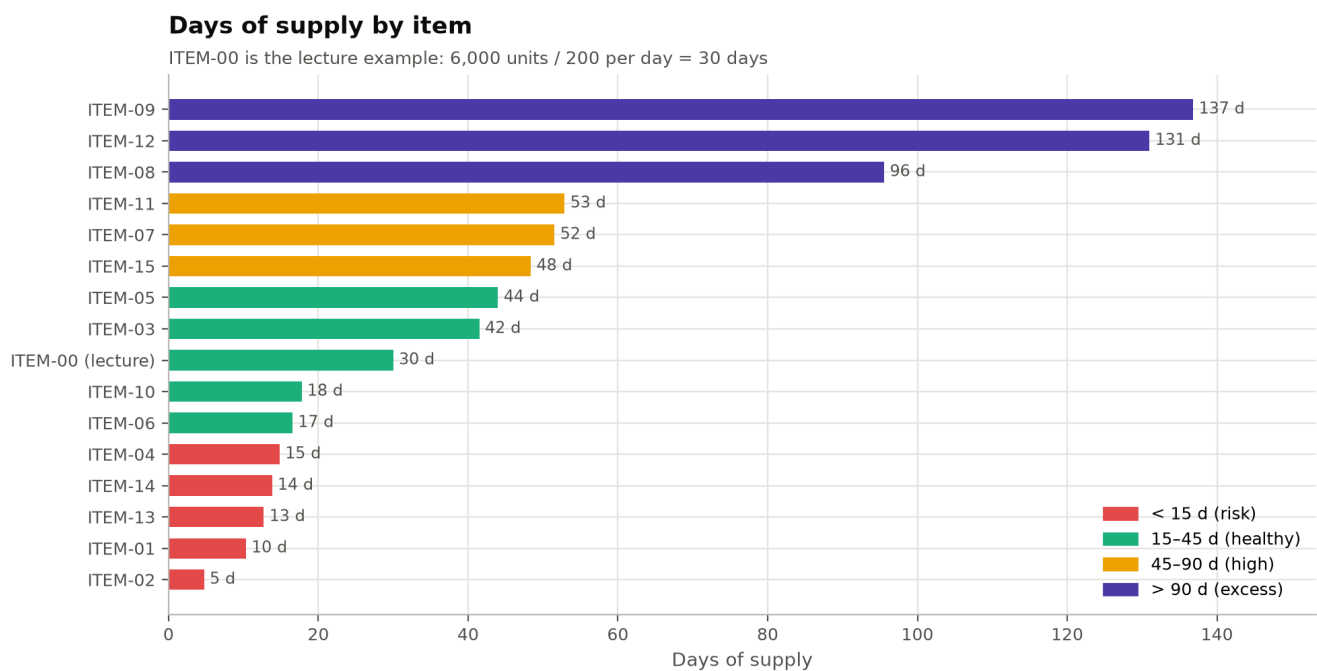


Figure 1.5 — Days of supply = inventory on hand / average daily usage.

Item days of supply and equivalent turns.

item	inventory on hand units	avg daily usage units	unit cost	days of supply	turns equiv	inventory value	band
ITEM-00 (lecture)	6,000	200	\$12.00	30.0	12.0	\$72,000	15–45 d (healthy)
ITEM-01	3,735	362	\$2.31	10.3	34.9	\$8,628	< 15 d (risk)
ITEM-02	1,343	283	\$4.28	4.7	75.9	\$5,748	< 15 d (risk)
ITEM-03	15,651	377	\$4.74	41.5	8.7	\$74,186	15–45 d (healthy)
ITEM-04	593	40	\$10.63	14.8	24.3	\$6,304	< 15 d (risk)
ITEM-05	11,396	259	\$4.93	44.0	8.2	\$56,182	15–45 d (healthy)
ITEM-06	2,662	161	\$4.54	16.5	21.8	\$12,085	15–45 d (healthy)
ITEM-07	7,834	152	\$8.66	51.5	7.0	\$67,842	45–90 d (high)
ITEM-08	13,666	143	\$5.94	95.6	3.8	\$81,176	> 90 d (excess)
ITEM-09	30,101	220	\$13.52	136.8	2.6	\$406,966	> 90 d (excess)
ITEM-10	373	21	\$18.65	17.8	20.3	\$6,956	15–45 d (healthy)
ITEM-11	13,644	258	\$7.92	52.9	6.8	\$108,060	45–90 d (high)
ITEM-12	13,491	103	\$3.24	131.0	2.7	\$43,711	> 90 d (excess)
ITEM-13	4,919	387	\$15.44	12.7	28.3	\$75,949	< 15 d (risk)
ITEM-14	3,550	257	\$8.87	13.8	26.1	\$31,488	< 15 d (risk)
ITEM-15	15,337	317	\$4.17	48.4	7.4	\$63,955	45–90 d (high)